Sapienza Università di Roma

Department of Mechanical and Aerospace Engineering

Homework 1 — Indirect Optimization of a Planar Ascent Trajectory

Dynamics and Control of Launch Vehicles

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Contents

1 Introduction

This report presents the solution to Homework 1 of the course *Dynamics and Control of Launch Vehicles*, which concerns the minimum-fuel planar ascent of a rocket from rest on the ground to a prescribed low-Earth orbit injection state. Four progressively richer mission programs are addressed, each framed as an optimal control problem and solved by applying the Pontryagin Maximum Principle (PMP) and single-shooting over the costate initial conditions.

1.1 Vehicle Dynamics

The vehicle is modelled as a point mass moving in a vertical plane above a flat, non-rotating Earth with no atmospheric drag. The state vector is

$$\mathbf{x}(t) = \big[x,\ y,\ v_x,\ v_y,\ m\big]^T, \tag{1}$$

where (x, y) is the position in the orbital plane, (v_x, v_y) is the velocity, and m is the vehicle mass. The single control variable is the thrust-direction angle φ measured from the horizontal. The thrust magnitude is $T = cQ$, where c is the effective exhaust velocity and $Q = -\dot{m}$ is the constant mass-flow rate. The equations of motion are

$$\dot{x} = v_x, \tag{2}$$
$$\dot{y} = v_y, \tag{3}$$
$$\dot{v}_x = \frac{T}{m}\cos\varphi, \tag{4}$$
$$\dot{v}_y = \frac{T}{m}\sin\varphi - g, \tag{5}$$
$$\dot{m} = -Q. \tag{6}$$

1.2 Non-dimensionalization

All quantities are made dimensionless with the reference values in Table 1. The reference velocity V_{ref} is the circular orbital speed at the nominal target altitude; the reference length and time follow from g.

Table 1: Non-dimensionalization reference values.

Quantity	Symbol	Expression	Value
Velocity	V_{ref}	orbital speed	7800 m/s
Acceleration	g_{ref}	g	9.81 m/s^2
Length	L_{ref}	V_{ref}^2/g	$\approx$ 6200 km
Time	t_{ref}	V_{ref}/g	$\approx$ 795 s
Mass	m_{ref}	m_0	10^6 kg

In non-dimensional form ($g = 1$, $m_0 = 1$, $v_{x,f} = 1$) the fixed problem data are collected in Table 2.

Table 2: Fixed non-dimensional problem parameters.

Symbol	Value	Physical interpretation
c	0.6	Effective exhaust velocity ($c_{\rm dim} \approx 4680$ m/s, $I_{sp} \approx 477$ s)
η	0.1	Structural coefficient m_s/m_p
y_f	0.04, 0.05, 0.06	Target orbit altitude ($\approx$ 248, 310, 372 km)

The mass-flow rate Q is a free design parameter swept in Tasks 1 and 4. Since $W = mg$ and $m_0 = 1$, the initial thrust-to-weight ratio is $T/W\big|_{t=0} = cQ$, so liftoff ($T > W$) requires $Q > 1/c \approx 1.67$. As propellant burns off ($\dot{m} = -Q < 0$) the ratio $T/W = cQ/m$ increases monotonically, hence the liftoff instant is the binding case.

1.3 Pontryagin Maximum Principle

1.3.1 Hamiltonian and Costate Equations

The optimal control problem is to find the control $\varphi(t)$ and the free terminal time t_f that maximize the final mass $m(t_f)$—equivalently, minimize the Mayer cost $J = -m(t_f)$—subject to the dynamics (2)–(6) and the terminal constraints

$$y(t_f) = y_f, \qquad v_x(t_f) = 1, \qquad v_y(t_f) = 0. \tag{7}$$

The horizontal position $x(t_f)$ is free, and t_f is free.

Introducing the costate (adjoint) vector $\boldsymbol{\lambda} = [\lambda_x,\ \lambda_y,\ \lambda_{v_x},\ \lambda_{v_y},\ \lambda_m]^T$, the Hamiltonian is

$$\mathcal{H} = \lambda_x v_x + \lambda_y v_y + \lambda_{v_x} \frac{T}{m} \cos\varphi + \lambda_{v_y} \left(\frac{T}{m} \sin\varphi - 1 \right) - \lambda_m Q. \tag{8}$$

Rearranging,

$$\mathcal{H} = \underbrace{\lambda_x v_x + \lambda_y v_y - \lambda_{v_y} - \lambda_m Q}_{\text{state-only part}} + \frac{T}{m} \underbrace{\left(\lambda_{v_x} \cos\varphi + \lambda_{v_y} \sin\varphi\right)}_{\text{control-dependent part}}. \tag{9}$$

The costate equations $\dot{\boldsymbol{\lambda}} = -\partial\mathcal{H}/\partial\mathbf{x}$ yield

$$\dot{\lambda}_x = 0 \qquad \Longrightarrow \qquad \lambda_x = 0 \quad (x(t_f) \text{ free } \Rightarrow \lambda_x(t_f) = 0), \tag{10}$$

$$\dot{\lambda}_y = 0 \qquad \Longrightarrow \qquad \lambda_y = \text{const}, \tag{11}$$

$$\dot{\lambda}_{v_x} = -\lambda_x = 0 \qquad \Longrightarrow \qquad \lambda_{v_x} = \lambda_{v_x,0} = \text{const}, \tag{12}$$

$$\dot{\lambda}_{v_y} = -\lambda_y \qquad \Longrightarrow \qquad \lambda_{v_y}(t) = \lambda_{v_y,0} - \lambda_y t, \tag{13}$$

$$\dot{\lambda}_m = \frac{T}{m^2} \left\| \boldsymbol{\lambda}_v \right\|, \tag{14}$$

where $\|\boldsymbol{\lambda}_v\| = \sqrt{\lambda_{v_x}^2 + \lambda_{v_y}^2}$.

Four of the five costate components are determined analytically: $\lambda_x = 0$ identically, λ_y and λ_{v_x} are constant, and λ_{v_y} varies linearly in time. Only λ_m requires numerical integration via Eq. (14). This analytical structure greatly reduces the cost of evaluating the shooting residual.

1.3.2 Optimal Control Law

By the PMP, the optimal control φ^* maximizes $\mathcal{H}$ pointwise. Since $T/m > 0$, this is equivalent to maximizing the control-dependent part of Eq. (8),

$$g(\varphi) = \lambda_{v_x}\cos\varphi + \lambda_{v_y}\sin\varphi, \tag{15}$$

over φ. The first-order stationarity condition reads

$$\frac{\partial\mathcal{H}}{\partial\varphi} = \frac{T}{m}\big(-\lambda_{v_x}\sin\varphi + \lambda_{v_y}\cos\varphi\big) = 0 \qquad \Longrightarrow \qquad \tan\varphi^* = \frac{\lambda_{v_y}(t)}{\lambda_{v_x}}. \tag{16}$$

This is the **bilinear-tangent (linear-tangent) law**: since λ_{v_y} is linear in t (Eq. 13) and λ_{v_x} is constant, $\tan\varphi^*$ varies linearly with time.

Geometric interpretation and branch selection. Equation (16) determines φ^* only up to an additive π: it admits two roots per period. Geometrically, λ_{v_x} and λ_{v_y} are the two legs of a right triangle whose hypotenuse is the norm $\|\boldsymbol{\lambda}_v\| = \sqrt{\lambda_{v_x}^2 + \lambda_{v_y}^2}$, so the two candidate directions are

$$\text{(I)}\quad \cos\varphi^* = \frac{\lambda_{v_x}}{\|\boldsymbol{\lambda}_v\|},\ \sin\varphi^* = \frac{\lambda_{v_y}}{\|\boldsymbol{\lambda}_v\|} \qquad \text{or} \qquad \text{(II)}\quad \cos\varphi^* = -\frac{\lambda_{v_x}}{\|\boldsymbol{\lambda}_v\|},\ \sin\varphi^* = -\frac{\lambda_{v_y}}{\|\boldsymbol{\lambda}_v\|}. \tag{17}$$

Substituting each branch into Eq. (15) discriminates between them:

$$g_{\mathrm{I}} = \frac{\lambda_{v_x}^2 + \lambda_{v_y}^2}{\|\boldsymbol{\lambda}_v\|} = \frac{\|\boldsymbol{\lambda}_v\|^2}{\|\boldsymbol{\lambda}_v\|} = +\|\boldsymbol{\lambda}_v\|, \qquad g_{\mathrm{II}} = -\|\boldsymbol{\lambda}_v\|. \tag{18}$$

Branch (I) yields the maximum and branch (II) the minimum, so the PMP selects branch (I): the optimal thrust direction is aligned with the velocity costate $\boldsymbol{\lambda}_v$ (primer vector), and the control-dependent term attains its largest value,

$$\lambda_{v_x}\cos\varphi^* + \lambda_{v_y}\sin\varphi^* = \|\boldsymbol{\lambda}_v\|. \tag{19}$$

Using this result, the optimal Hamiltonian simplifies to

$$\mathcal{H}^* = \lambda_y v_y + \frac{T}{m}\|\boldsymbol{\lambda}_v\| - \lambda_{v_y} - \lambda_m Q. \tag{20}$$

1.3.3 Transversality Conditions

The costate boundary conditions follow from the transversality (natural boundary) conditions of the PMP. Adjoining the terminal constraints $\boldsymbol{\psi}(\mathbf{x}(t_f)) = [\,y - y_f,\ v_x -$

$1,\ v_y]^T = \mathbf{0}$ with multipliers $\boldsymbol{\nu}$ and the dynamics with the costates $\boldsymbol{\lambda}$, the first variation of the augmented payoff $\Phi = m(t_f)$ produces a boundary term at t_f that vanishes only if

$$\boldsymbol{\lambda}(t_f) = \frac{\partial \Phi}{\partial \mathbf{x}_f} + \left(\frac{\partial \boldsymbol{\psi}}{\partial \mathbf{x}_f}\right)^T \boldsymbol{\nu}. \tag{21}$$

Each terminal state falls into one of three categories. The *constrained* states y, v_x, v_y leave their costates $\lambda_y(t_f), \lambda_{v_x}(t_f), \lambda_{v_y}(t_f)$ undetermined: these absorb the unknown multipliers $\boldsymbol{\nu}$ and are carried as shooting unknowns (Eq. 24). The horizontal position x is *free and absent from the payoff*, so $\partial\Phi/\partial x = 0$ and Eq. (21) returns $\lambda_x(t_f) = 0$, consistent with $\lambda_x \equiv 0$ already obtained in Eq. (10). Finally, the terminal mass $m(t_f)$ is *free but present in the payoff*, with $\partial\Phi/\partial m = 1$, which fixes the only nontrivial terminal costate:

$$\lambda_m(t_f) = \frac{\partial \Phi}{\partial m(t_f)} = 1. \tag{22}$$

This condition carries a twofold meaning. First, $\lambda_m(t_f)$ is the sensitivity of the optimal payoff to the terminal mass (a unit increase in $m(t_f)$ improves the objective by one unit). Second, since the PMP determines $\boldsymbol{\lambda}$ only up to a positive scale factor—the Hamiltonian is homogeneous of degree one in $\boldsymbol{\lambda}$—Eq. (22) fixes the normalization of the costate vector.[1]

Because the terminal time t_f is free, the PMP supplies one further condition—the natural boundary condition for t_f:

$$\mathcal{H}(t_f) = 0. \tag{23}$$

The system is autonomous (no explicit time dependence), so the Hamiltonian is constant along an optimal trajectory; $\mathcal{H}(t_f) = 0$ therefore implies $\mathcal{H}(t) \equiv 0$ throughout the powered arc, consistent with the optimality conditions.

1.4 Single-Shooting Method and Continuation

Substituting the bilinear-tangent law (16) into Eqs. (2)–(6), the optimal control problem reduces to a multi-point **Boundary Value Problem (BVP)**. The shooting unknowns common to all four tasks are

$$\boldsymbol{\zeta} = \left[\lambda_{v_x,0},\ \lambda_{v_y,0},\ \lambda_y,\ \lambda_{m,0},\ t_f\right]^T. \tag{24}$$

Given a trial $\boldsymbol{\zeta}$, the augmented state $[x, y, v_x, v_y, m, \lambda_m]^T$ is integrated forward with `ode45` (tolerances: RelTol $= 10^{-10}$, AbsTol $= 10^{-12}$), and the residual vector is

[1] The sign depends on the chosen convention. Maximizing the payoff $\Phi = m(t_f)$ yields $\lambda_m(t_f) = +1$, consistent with the maximization form of $\mathcal{H}$ adopted here and with $\dot{\lambda}_m = (T/m^2)\|\boldsymbol{\lambda}_v\| > 0$ along the arc. The equivalent minimization of $J = -m(t_f)$ flips the costate sign convention and recovers the same physical solution.

evaluated at t_f. The system is solved with `fsolve` (function and step tolerances both 10^{-10}, up to 500 iterations).

Single-shooting is sensitive to the costate initial guess, which has no direct physical interpretation. The **continuation strategy** mitigates this: the BVP is first solved at a favorable operating point ($Q \approx 3$), and each subsequent value of Q is initialized with the converged solution from the neighboring point, sweeping both forward and backward across the range $Q \in [1.8, 7]$. This warm-start approach propagates the solution branch smoothly without manual retuning of the initial guess.

2 Task 1: Single Burn Arc

2.1 Problem Statement

The simplest mission profile consists of a single continuous powered arc starting from rest at the origin, $(x, y, v_x, v_y) = (0, 0, 0, 0)$ with $m(0) = 1$, and ending at the orbital injection state satisfying Eq. (7). The mass-flow rate Q is the primary free parameter swept over the range $Q \in [1.8, 7]$; values below $Q_{\min} = 1/c \approx 1.67$ would yield insufficient thrust at liftoff ($T/W\big|_{t=0} = cQ < 1$).

2.2 Boundary Value Problem Formulation

The five shooting unknowns $\boldsymbol{\zeta}$ (Eq. 24) determine both the costate initial conditions and the free flight time. The shooting residual is

$$\mathbf{r}_1(\boldsymbol{\zeta}) = \begin{bmatrix} y(t_f) - y_f \\ v_x(t_f) - 1 \\ v_y(t_f) \\ \lambda_m(t_f) - 1 \\ \mathcal{H}(t_f) \end{bmatrix} = \mathbf{0}, \tag{25}$$

where $\mathcal{H}(t_f)$ is computed via Eq. (20). The first three components enforce the terminal state constraints (7); the fourth is the transversality condition (22) on the mass costate; the fifth is the free-time condition (23).

2.3 Velocity-Loss Decomposition

The total velocity loss $W_{\text{tot}} = \Delta V_{\text{ideal}} - v_f$ compares the ideal Tsiolkovsky delta-V,

$$\Delta V_{\text{ideal}} = c \ln \frac{m_0}{m_f} = c \ln \frac{1}{m_f}, \tag{26}$$

against the achieved injection speed $v_f = 1$. It is decomposed as

$$W_d = \int_0^{t_f} \frac{T}{m} \big[1 - \cos(\varphi - \psi)\big] \, \mathrm{d}t \qquad \text{(thrust misalignment loss)}, \tag{27}$$

$$W_g = \int_0^{t_f} \sin \psi \, \mathrm{d}t \qquad \text{(gravity loss)}, \tag{28}$$

where $\psi = \arctan(v_y/v_x)$ is the flight-path angle. Misalignment loss W_d arises from the fraction of thrust not aligned with the velocity vector; gravity loss W_g is the work done against gravity projected along the trajectory.

2.4 Results

Figure 1 shows the final mass as a function of Q for the three target altitudes. Each curve exhibits a clear interior maximum Q^*: at low Q, the long burn time accumulates large gravity losses; at high Q, the vehicle climbs steeply and thrust-misalignment loss dominates.

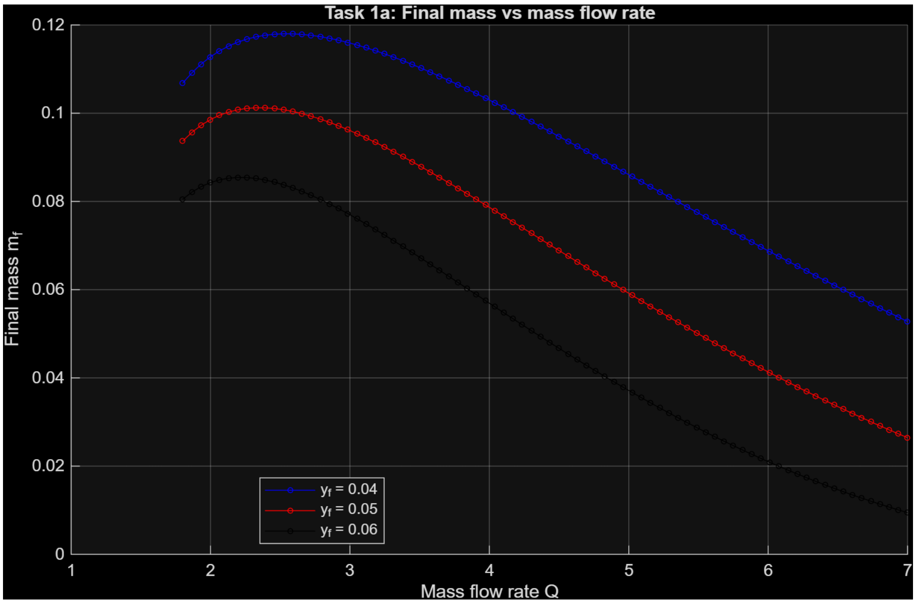


Figure 1: Task 1a: final mass m_f as a function of the mass-flow rate Q for three target altitudes y_f.

Figure 2 confirms this trade-off by decomposing the velocity loss at $y_f = 0.04$. The gravity loss W_g is large at low Q and decreases monotonically with Q, while the misalignment loss W_d has a broad minimum near Q^*.

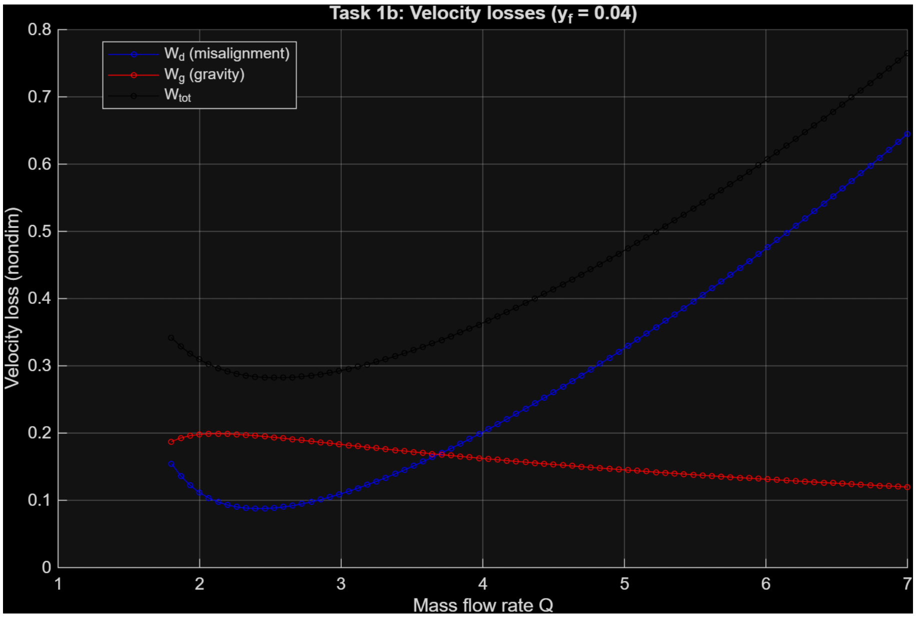
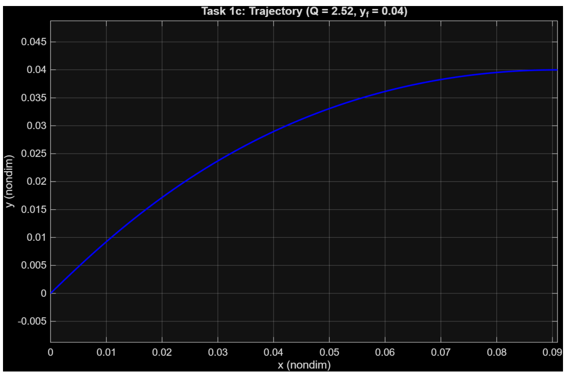
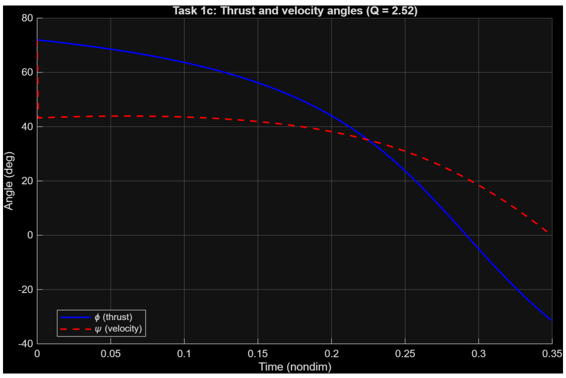


Figure 2: Task 1b: velocity-loss decomposition—misalignment W_d, gravity W_g, and total W_{tot}—at $y_f = 0.04$.

Figures 3a and 3b show the optimal trajectory and control history at $Q = Q^*$ for $y_f = 0.04$. The thrust angle φ decreases smoothly from near-vertical at launch to near-horizontal at injection, following the bilinear-tangent law (16). The velocity angle ψ tracks φ closely once the initial transient has died out.



(a) Optimal trajectory in the (x, y) plane.



(b) Thrust angle φ and velocity angle ψ vs. time.

Figure 3: Task 1c: optimal trajectory and control at $Q = Q^*$, $y_f = 0.04$.

3 Task 2: Vertical Climb and Optimal Burn

3.1 Phase 1: Vertical Climb

Practical launchers cannot pitch over immediately at liftoff because of aerodynamic loads and tower-clearance requirements. Task 2 prepends a short *vertical-climb phase* ($\varphi = \pi/2$, thrust pointing straight up) until the altitude reaches $y_1 = 10^{-4}$ (dimensionally ≈ 620 m). During this phase the horizontal position and velocity remain zero, and the 3-state sub-problem is

$$\dot{y} = v_y, \qquad \dot{v}_y = \frac{T}{m} - 1, \qquad \dot{m} = -Q, \tag{29}$$

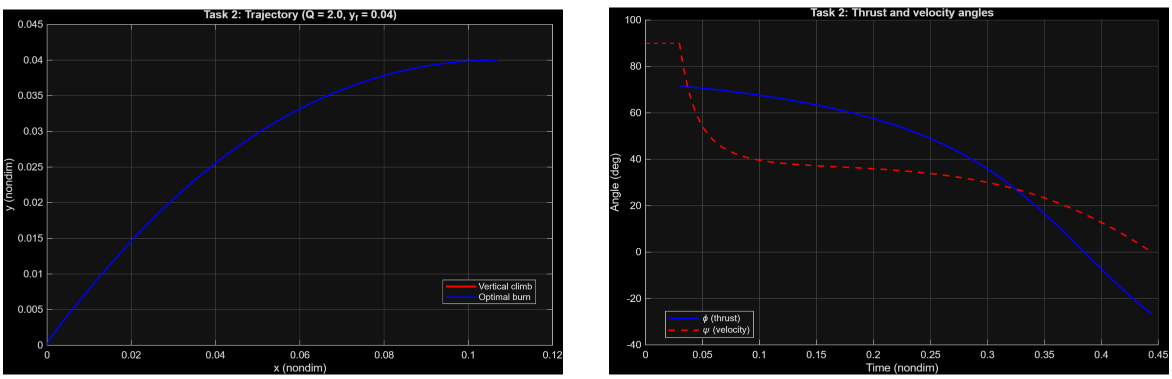
integrated until the event $y = y_1$ is detected via `ode45` event handling. The end state $(y_1,\ v_{y,1},\ m_1)$ becomes the initial condition for Phase 2.

3.2 Phase 2: Optimal Burn

Starting from the state $(x_0, y_0, v_{x,0}, v_{y,0}, m_0) = (0,\ y_1,\ 0,\ v_{y,1},\ m_1)$, Phase 2 is an optimal powered arc with the same BVP structure as Task 1. The shooting unknowns and the residual (25) are unchanged; only the initial conditions of the augmented state differ. Because the orbit injection targets are identical to Task 1, the continuation warm-start from Task 1's solution at $Q = 2$ provides a reliable initial guess.

3.3 Results

Figure 4 shows the composite trajectory and the angle history for $Q = 2$, $y_f = 0.04$. The red segment is the short vertical-climb phase; the blue arc is the optimal burn. The trajectory is qualitatively similar to Task 1 except for the initial vertical stub. During the climb both φ and ψ are fixed at 90°; at the transition the thrust begins to rotate toward the horizontal following the bilinear-tangent law, while the velocity angle ψ lags momentarily before catching up as horizontal speed builds.



(a) Composite trajectory: vertical climb (red), optimal burn (blue).

(b) Thrust angle φ and velocity angle ψ during the burn.

Figure 4: Task 2: trajectory and control history for $Q = 2$, $y_f = 0.04$.

The final mass in Task 2 is marginally lower than in Task 1: the vertical climb consumes propellant while building only vertical velocity, contributing to gravity losses without advancing the vehicle toward the horizontal injection speed. Nevertheless, the prepended climb is a standard launch practice, and the payload penalty is modest for the short y_1 value used here.

4 Task 3: Vertical Climb, Burn, and Coast

4.1 Switching Function and Thrust-Arc Structure

Task 3 introduces a *coast arc* between the powered burn and orbital injection. In the PMP framework, the switching function governs whether the engine is on or off:

$$S(t) = \frac{\|\boldsymbol{\lambda}_v(t)\|}{m(t)} - \frac{1}{c}. \tag{30}$$

The engine is *on* (full thrust) when $S > 0$ and *off* (coast) when $S < 0$. During a coast arc $T = 0$, so $\dot{m} = 0$ and the vehicle follows purely ballistic dynamics.

The three-phase mission sequence is:

1. **Vertical climb**: $\varphi = \pi/2$ until $y = y_1 = 10^{-4}$ (identical to Task 2),
2. **Powered burn**: bilinear-tangent steering until engine cutoff at time t_c,
3. **Coast arc**: ballistic flight from t_c to the injection point.

4.2 Engine Cutoff Conditions

The shooting unknowns remain $\boldsymbol{\zeta} = [\lambda_{v_x,0},\ \lambda_{v_y,0},\ \lambda_y,\ \lambda_{m,0},\ t_{\text{burn}}]^T$ (with time measured from the start of Phase 2). At engine cutoff the following five conditions must be satisfied simultaneously:

$$\mathbf{r}_3(\boldsymbol{\zeta}) = \begin{bmatrix} v_{x,c} - 1 \\ y_c + \frac{1}{2}v_{y,c}^2 - y_f \\ \lambda_m(t_c) - 1 \\ \lambda_{v_y}(t_c) - \lambda_y\, v_{y,c} \\ S(t_c) \end{bmatrix} = \mathbf{0}. \tag{31}$$

Conditions 1 and 2 encode the *ballistic-coast reachability*: since during the coast v_x is constant and v_y decreases linearly under gravity, the vehicle must already have the orbital horizontal speed ($v_{x,c} = 1$) at cutoff, and the apogee of the ballistic arc must coincide with the target altitude. Condition 3 is the standard mass-costate transversality (22). Condition 4 ensures $\lambda_{v_y}(t_f) = 0$ at the coast end: since λ_{v_y} evolves linearly throughout, $\lambda_{v_y}(t_f) = \lambda_{v_y}(t_c) - \lambda_y v_{y,c}$, and setting this to zero recovers condition 4. Condition 5 is the bang-bang switching condition: the engine cuts off exactly when $S = 0$.

4.3 Coast Arc

During the ballistic coast ($T = 0$) the trajectory is analytically integrable. Denoting the elapsed coast time by τ and the cutoff state by subscript c:

$$x(t_c + \tau) = x_c + v_{x,c}\,\tau, \qquad y(t_c + \tau) = y_c + v_{y,c}\,\tau - \tfrac{1}{2}\tau^2. \tag{32}$$

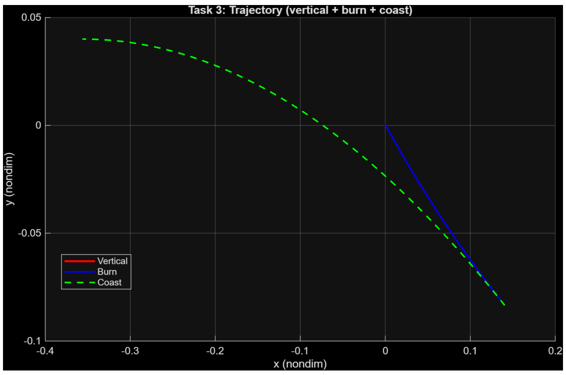
The coast ends at apogee $\tau^* = v_{y,c}$, where $v_y = 0$ and

$$y_f = y_c + \tfrac{1}{2}v_{y,c}^2, \tag{33}$$

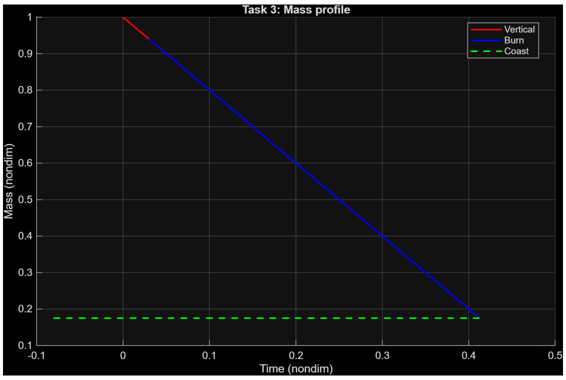
in agreement with condition 2 of (31). The vehicle mass is constant throughout the coast: $m_f = m_c$.

4.4 Results

Figure 5 shows the three-phase trajectory and mass profile for $Q = 2$, $y_f = 0.04$. The vertical climb (red) is short; the burn arc (blue) terminates below the target altitude; the ballistic coast (green dashed) ascends to the injection point. The mass profile confirms the model: it decreases linearly during both powered phases and remains constant during the coast. The final mass in Task 3 exceeds that of Task 2: by cutting the engine early and relying on the ballistic coast, less propellant is consumed while still satisfying all terminal constraints.



(a) Trajectory: vertical climb (red), burn (blue), coast (green dashed).



(b) Mass profile: linear decrease during powered phases, flat during coast.

Figure 5: Task 3: trajectory and mass profile for $Q = 2$, $y_f = 0.04$.

Figure 6 shows the thrust and velocity angles throughout the three phases. The dashed vertical lines mark the end of the vertical climb and the engine cutoff. During the coast no thrust angle is defined; the velocity angle ψ continues to evolve ballistically until it reaches zero at apogee.

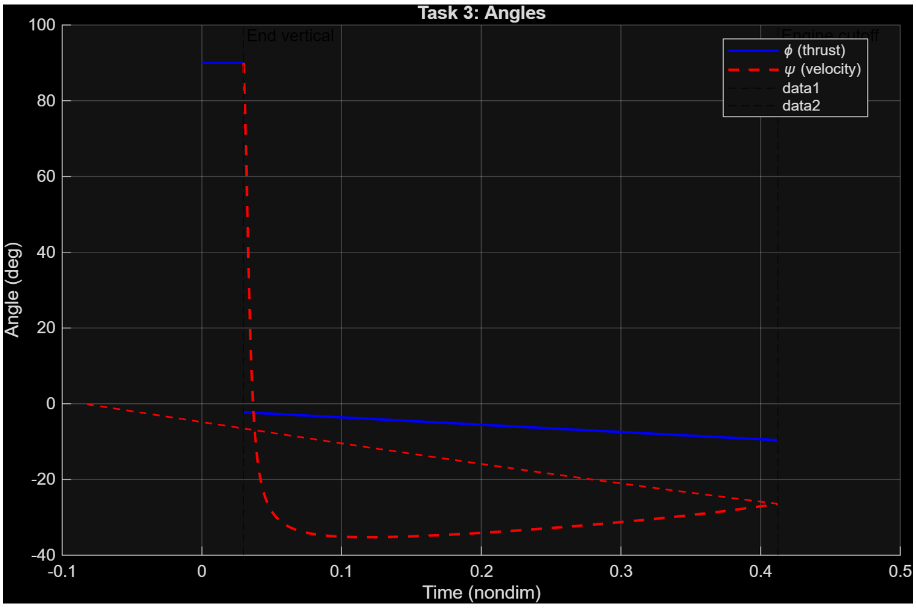


Figure 6: Task 3: thrust angle φ and velocity angle ψ across all three phases. Vertical dashed lines mark phase transitions.

5 Task 4: Optimal Staging

5.1 Staging Model

Task 4 removes the vertical-climb phase and introduces a single *staging event* at time $t = t_s$. At staging, the first-stage structural mass

$$m_{s1} = \eta\, Q\, t_s \tag{34}$$

is jettisoned instantaneously, where $\eta\, Q\, t_s$ is the structural fraction of the propellant consumed by the first stage. The vehicle mass undergoes a step:

$$m^+(t_s) = m^-(t_s) - m_{s1} = 1 - Q\, t_s - \eta\, Q\, t_s = 1 - Q\, t_s(1+\eta). \tag{35}$$

For $m^+(t_s) > 0$ one requires $t_s < [Q(1+\eta)]^{-1}$, which provides an upper bound on the admissible staging time. The costate vector is continuous across the staging event.

The net payload delivered to orbit is

$$m_{\text{pay}} = m_f - m_{s2}, \qquad m_{s2} = \eta\, Q\,(t_f - t_s), \tag{36}$$

where m_{s2} is the second-stage structural mass (proportional to the second-stage propellant consumption).

5.2 Optimization Procedure

For a fixed staging time t_s, the two burn arcs share the same costate evolution (the bilinear-tangent law is unaffected by staging), and the BVP reduces to the same structure as Task 1 with updated initial mass after staging:

1. Integrate the augmented state from $t = 0$ to $t = t_s$ (Stage 1 burn),
2. Apply the staging mass jump (34),
3. Integrate from t_s to t_f (Stage 2 burn),
4. Evaluate the residual (25) at t_f.

The shooting unknowns $\boldsymbol{\zeta}$ (Eq. 24) are identical to Task 1.

The optimal staging time t_s^* is found by an **outer parametric sweep**: t_s is varied over a dense grid of 50 points, the inner BVP is solved at each point (warm-started from the previous converged solution via continuation), and the payload (36) is maximized over the converged solutions. The starting guess for the inner BVP at the first grid point uses the single-stage solution from Task 1.

5.3 Results

Figure 7 shows the payload as a function of staging time t_s for $Q = 2$, $y_f = 0.04$. The curve has a well-defined maximum; the single-stage baseline (dashed red) lies significantly below the two-stage peak.

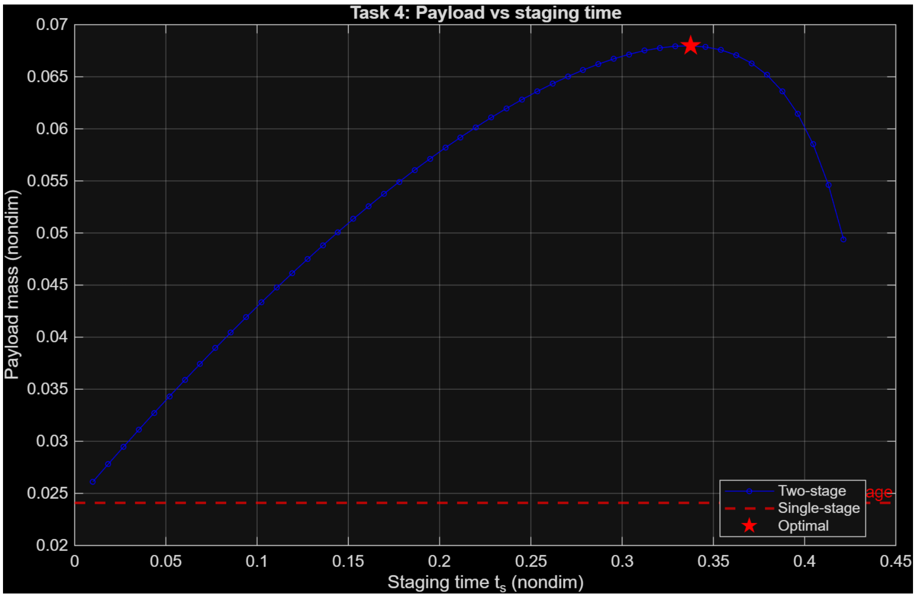


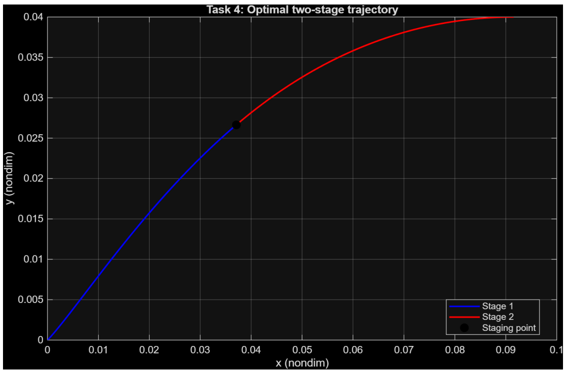
Figure 7: Task 4: payload mass vs. staging time t_s. Red dashed line: single-stage reference. Red star: optimal staging point.

Table 3 summarises the numerical results at $Q = 2$, $y_f = 0.04$.

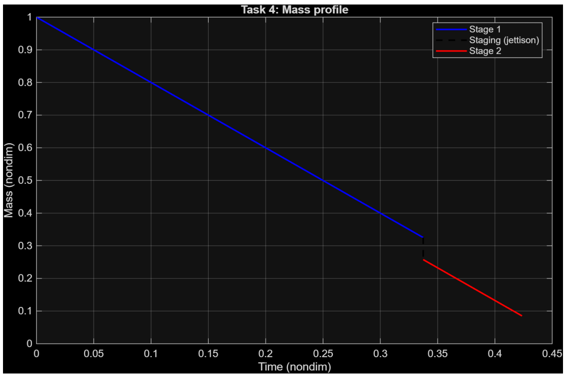
Table 3: Task 4 key numerical results ($Q = 2$, $y_f = 0.04$).

Quantity	Value	Notes
Optimal staging time t_s^*	0.337	nondim ($\approx$ 268 s)
Single-stage payload	0.0241	nondim
Two-stage payload	0.0680	nondim
Payload gain	+182%	vs. single-stage

Figures 8a and 8b show the optimal two-stage trajectory and mass profile. The discontinuous mass drop at t_s^* is visible in the profile; the trajectory is continuous (position and velocity are unaffected by staging, only mass changes).



(a) Optimal two-stage trajectory: Stage 1 (blue), Stage 2 (red), staging point (circle).



(b) Mass profile with the instantaneous jettison at t_s^*.

Figure 8: Task 4: optimal two-stage solution for $Q = 2$, $y_f = 0.04$.

The large relative gain is a consequence of the model simplifications (constant Q, no drag, identical I_{sp} for both stages). Real launchers achieve much smaller but still substantial staging benefits; the key mechanism—jettisoning inert structural mass before it has to be accelerated to orbital velocity—is the same.

6 Conclusions

This work applied the Pontryagin Maximum Principle to four minimum-fuel planar ascent programs of increasing complexity, all solved by single-shooting with `fsolve` over the costate initial conditions. The main findings are as follows.

Bilinear-tangent law (all tasks). The flat-Earth, drag-free dynamics produce a particularly tractable adjoint structure: four of the five costate components are determined analytically (two constants and one linear ramp), leaving only the mass costate λ_m to be integrated numerically. The resulting bilinear-tangent steering law reduces the optimal control problem to a five-dimensional root-finding problem, making single-shooting computationally affordable.

Optimal thrust-to-weight ratio (Task 1). The mass-flow rate Q that maximises the final mass balances two competing losses: gravity losses (dominant at low Q, due to long burn times) and thrust-misalignment losses (dominant at high Q, due to steep climb angles). The optimal Q^* shifts slightly with target altitude y_f, reflecting the longer burn time required for higher orbits.

Continuation is indispensable. Single-shooting from a cold start fails across most of the Q range. The continuation strategy—warm-starting each new Q from the previous converged solution and sweeping forward and backward from a seed point—was the critical algorithmic ingredient enabling the parameter sweeps in Tasks 1 and 4.

Coast arc saves propellant (Task 3). Terminating the engine before reaching the target altitude and relying on ballistic dynamics is beneficial whenever the vehicle has sufficient residual vertical velocity to coast up to the injection point. The condition $S(t_c) = 0$ determines the optimal engine cutoff time without an additional outer optimization loop.

Staging provides large payload gains (Task 4). At the reference operating point ($Q = 2$, $y_f = 0.04$), the optimal two-stage configuration delivers a +182% payload gain over the single-stage baseline. The fundamental mechanism—discarding first-stage structural mass before it needs to be accelerated to orbital velocity—is well-captured even by the simplified constant-Q, drag-free model.

Model limitations. The simplifications adopted throughout (flat Earth, no drag, constant thrust, no vehicle attitude dynamics) neglect key physical effects. Atmospheric drag dominates energy loss in the first 80 km of ascent; Earth's curvature and rotation alter the optimal coast and staging conditions; and real throttling constraints bound Q away from its theoretical range. Extending the framework to 3-DoF spherical-Earth dynamics with a drag model would shift Q^* and t_s^* quantitatively but would not alter the PMP structure derived here.

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